The success of Zipline highlights the critical role of policy support, coordinated infrastructure planning, and local healthcare workforce collaboration in enabling UAV logistics at national scale. It also illustrates how drone-enabled medical supply chains can improve equity in healthcare access, strengthen national resilience during emergencies, and offer a model for other countries seeking to modernize essential service delivery.

8.10.7 C ase study: Amazon Prime Air – commercial parcel delivery in the United States

Amazon Prime Air represents one of the most notable efforts to incorporate UAVs into commercial last-mile delivery networks. Announced in 2013 and refined through multiple phases of testing, Prime Air aims to deliver packages weighing up to approximately 2.5 kilograms directly to customers within 30 minutes of ordering. This initiative is driven by Amazon's broader logistics strategy of achieving ultra-fast, on-demand delivery while reducing reliance on traditional road-based courier networks, which are increasingly affected by traffic congestion, fuel costs, and environmental sustainability concerns.

Prime Air utilizes autonomous multirotor delivery drones equipped with computer vision, onboard collision avoidance systems, object-recognition sensors, and encrypted navigation software. These drones operate from local fulfilment nodes and fly directly to the delivery destination, where they safely lower packages into designated landing zones near the recipient's home. The drones maintain strict geofenced flight corridors and are programed to detect obstacles such as trees, wires, buildings, and moving objects. Several trial deployments have been launched in California, Texas, and select regions in the United Kingdom, under Federal Aviation Administration (FAA) and Civil Aviation Authority (CAA) regulatory oversight (Welch, 2015).